A REPORT

ON

Cybersecurity Internship at InternSpark

Submitted by

Mr. Prince Kumar - 20241CSE0194

Under the guidance of,

Ms. Akkamaha Devi
Mr. Vishnu Shankar

in partial fulfillment for the award of the

degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING (COMPUTER ENGINEERING)

AT

GAIN MORE KNOWLEDGE
REACH GREATER HEIGHTS

PRESIDENCY UNIVERSITY

BENGALURU

AUGUST 2026

PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

GAIN MORE KNOWLEDGE
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CERTIFICATE

This is to certify that the report of **CSE7000 – Internship** on **"Cybersecurity Internship at InternSpark"** being submitted by "**Prince Kumar**" bearing roll number "20241CSE0194" in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in **Computer Science and Engineering (Computer Science and Engineering)** is a bonafide work carried out under our supervision.

Ms. Akkamaha Devi
PSCS
Presidency University

Mr. Vishnu Shankar
PSCS
Presidency University

Ms. Vidhya Rengasamy
Internship coordinator
PSCS
Presidency University

Dr. Bhuvaneshwari Patil
School level Internship coordinator
PSCS
Presidency University

Dr. Nagaraja S R
Head of the Department
PSCS
Presidency University

Dr. Duraipandian N
Dean – PSCS & PSIS
Presidency University

PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

I hereby declare that the work, which is being presented in the report entitled "**Cybersecurity Internship at InternSpark**" in partial fulfillment for the award of Degree of **Bachelor of Technology** in **Computer Science and Engineering (Computer Science and Engineering),** is a record of my own investigations carried under the guidance of Ms. Akkamaha Devi and Mr. Vishnu Shankar, Presidency School of Computer Science and Engineering, Presidency University, Bengaluru.

I have not submitted the matter presented in this report anywhere for the award of any other Degree.

Prince Kumar
20241CSE0194

INTERNSHIP COMPLETION CERTIFICATE

internSpark

All India Council for Technical Education
AICTE

CERTIFICATE
OF COMPLETION

THIS CERTIFICATE IS PRESENTED TO

Prince Kumar

In recognition of his/her efforts and achievements in completing **1 Month** internship in **Cybersecurity.**

Conducted From 5th June 2026 to 5th July 2026

ALFIDO TECH
RIDING THE WAVES OF TECHNOLOGICAL ADVANCEMENT

Candidate ID
IS-2026-9871

Scan to Verify Certificate Authenticity

ISO 9001:2015

MSME

InternSpark

Authorized Signatory

Issued by Alfido Tech – Registered Training Organization

ACKNOWLEDGEMENTS

First of all, I indebted to the **GOD ALMIGHTY** for giving me an opportunity to excel in our efforts to complete this internship on time.

I express sincere thanks to our respected Dean **Dr. Duraipandian N,** Presidency School of Computer Science and Engineering & Presidency School of Information Science, Presidency University for getting us permission to undergo the internship.

I express heartfelt gratitude to our beloved **Dr. Nagaraja S R**, Head of the Department, Presidency School of Computer Science and Engineering, Presidency University, for rendering timely help in completing this internship successfully.

I am greatly indebted to my reviewers **Ms. Akkamaha Devi** and **Mr. Vishnu Shankar,** Presidency School of Computer Science and Engineering, Presidency University for their inspirational guidance, and valuable suggestions and for providing a chance to express technical capabilities in every respect for the completion of the internship work.

I would like to convey gratitude and heartfelt thanks to the Internship Coordinator **Dr. Bhuvaneshwari Patil** and Program Internship Coordinator **Ms. Vidhya Rengasamy.** I thank faculty and staff members of Presidency University for their support during my Internship. And also I thank my family and friends for the strong support and inspiration they have provided us in bringing out this internship.

Prince Kumar

20241CSE0194

ABSTRACT

The Cybersecurity Internship at InternSpark was conducted from 5th June 2026 to 5th July 2026 with the objective of providing practical exposure to the field of cybersecurity through industry-oriented training and real-world assignments. During this internship, I acquired theoretical knowledge as well as practical skills related to various cybersecurity domains, including networking fundamentals, cyber threats, encryption techniques, vulnerability assessment, penetration testing, firewalls, ethical hacking, incident response, Security Operations Center (SOC) operations, and cyber laws.

The internship followed a structured learning approach where every topic was supported by practical assignments and assessments. The learning process began with the fundamentals of cybersecurity and computer networks, which provided a strong foundation for understanding how modern communication systems function. Gradually, advanced topics such as ethical hacking, penetration testing, and incident response were introduced to help understand how organizations identify, analyze, and respond to cyber threats.

Throughout the internship, multiple hands-on tasks helped in understanding the importance of securing computer systems and networks against various attacks. Different types of cyber attacks such as phishing, malware, ransomware, denial-of-service attacks, SQL injection, and password attacks were studied along with their preventive measures. Practical exercises on vulnerability assessment enabled me to identify weaknesses in systems and understand methods used to improve their security.

Another important aspect of the internship was learning about cryptography and encryption techniques that protect sensitive information during storage and transmission. Firewall technologies and Security Operations Center (SOC) activities demonstrated how organizations continuously monitor

networks to detect and respond to suspicious activities.

The internship also emphasized professional development through regular assignments, performance evaluations, and technical assessments. These activities enhanced my analytical thinking, problem-solving ability, communication skills, and teamwork. I successfully completed all assigned tasks and achieved an overall performance of 91.65% with Grade A+, demonstrating consistent dedication and understanding of cybersecurity concepts.

Overall, this internship strengthened my technical foundation in cybersecurity and increased my confidence in applying security concepts to real-world situations. It has motivated me to pursue advanced learning in information security and prepare for future roles in cybersecurity and ethical hacking.

INTRODUCTION

Cybersecurity has become one of the most critical fields in information technology due to the rapid growth of digital transformation and internet-based services. Organizations across the world face increasing cyber threats that target sensitive information, financial systems, communication networks, and critical infrastructure. Protecting digital assets requires skilled cybersecurity professionals capable of identifying vulnerabilities, preventing attacks, and responding effectively to security incidents.

The one-month internship at InternSpark was designed to bridge the gap between academic learning and industrial practices by providing structured training in cybersecurity concepts. During this internship, I gained practical exposure to industry-standard methodologies used to protect computer systems and networks. The internship focused on theoretical learning combined with practical assignments that simulated real-world cybersecurity scenarios.

The program introduced various cybersecurity concepts ranging from networking fundamentals to ethical hacking and incident response. Each module contributed towards developing practical skills and improving understanding of security principles used in organizations.

This internship significantly enhanced my technical knowledge and helped me understand the importance of cybersecurity in protecting modern digital infrastructure.

OBJECTIVES

The primary objectives of this internship were:

- To understand the basic principles of cybersecurity.
- To learn networking concepts and communication protocols.
- To identify different types of cyber threats and attacks.
- To study encryption techniques used for securing information.
- To perform vulnerability assessment using industry approaches.
- To understand penetration testing methodologies.
- To learn firewall configuration and network security.
- To study ethical hacking principles and responsible disclosure.
- To understand Security Operations Center (SOC) processes.
- To learn incident response procedures.
- To understand cyber laws and information security regulations.
- To improve technical documentation and report-writing skills.
- To enhance analytical thinking and problem-solving abilities.

ORGANIZATION PROFILE

InternSpark is an industry-oriented internship and training organization that provides students with practical exposure to emerging technologies through structured learning programs. The organization focuses on developing technical competencies by combining theoretical concepts with project-based assignments and continuous assessments.

The Cybersecurity Internship Program is designed to introduce students to modern security practices used in industries. The curriculum covers networking fundamentals, cyber threats, vulnerability assessment, ethical hacking, penetration testing, encryption, firewalls, incident response, and cyber laws.

Throughout the internship, experienced mentors guided students through practical exercises, assessments, and project-based learning activities. The organization emphasizes skill development, professional ethics, teamwork, and continuous learning.

TECHNOLOGIES AND TOOLS USED

During the internship, various cybersecurity concepts and tools were studied.

Networking

Understanding IP addressing, routing, switching, DNS, DHCP, HTTP, HTTPS, FTP, and TCP/IP communication.

Encryption

Learning symmetric and asymmetric encryption, hashing algorithms, digital signatures, SSL/TLS, and secure communication.

Vulnerability Assessment

Understanding system weaknesses and identifying security vulnerabilities through structured analysis.

Penetration Testing

Learning ethical methods used to identify security flaws before attackers exploit them.

Firewalls

Studying packet filtering, stateful inspection, proxy firewalls, and network access control.

Ethical Hacking

Understanding legal hacking techniques used for identifying vulnerabilities in systems.

Incident Response

Learning incident detection, containment, eradication, recovery, and post-incident analysis.

SOC Operations

Understanding continuous monitoring, log analysis, threat detection, and security event management.

Cyber Laws

Learning legal frameworks related to information security, cybercrime, privacy, and digital evidence.

WEEK-WISE WORK PROGRESS

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Week 1

- Introduction to Cybersecurity
- Networking Basics
- Types of Cyber Threats
- Encryption Techniques
- Vulnerability Assessment
- Penetration Testing
- Firewalls

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Week 2

- Ethical Hacking
- Incident Response
- SOC Operations
- Cyber Laws & Compliance
- Revision of Cybersecurity Concepts

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Week 3

- Advanced Encryption
- Vulnerability Assessment
- Penetration Testing
- Firewall Configuration
- Ethical Hacking
- Incident Response
- SOC Operations

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Week 4

- Cyber Laws
- Networking
- Encryption
- Threat Analysis
- Vulnerability Assessment
- Penetration Testing

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Week 5

- Firewall Management
- Ethical Hacking
- Final Incident Response Assessment

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CHALLENGES FACED

During the internship, understanding advanced cybersecurity concepts such as penetration testing, encryption algorithms, and vulnerability assessment required continuous practice. Initially, distinguishing between different attack methodologies and selecting appropriate defensive techniques was challenging. Time management while completing multiple assignments and assessments also required discipline. These challenges were overcome through regular practice, mentor guidance, and reviewing technical documentation.

RESULTS AND LEARNING OUTCOMES

The internship significantly improved my understanding of cybersecurity concepts and their practical applications. I developed skills in identifying cyber threats, analysing vulnerabilities, implementing basic security measures, and understanding organizational security practices. My communication, documentation, and technical reporting skills also improved. I successfully completed the internship with an **overall score of 91.65% and Grade A+**, reflecting consistent performance and active participation.

CONCLUSION

The Cybersecurity Internship at InternSpark provided valuable exposure to the practical aspects of information security and cyber defense. The structured learning approach, combined with hands-on assignments and assessments, helped bridge the gap between classroom knowledge and real-world cybersecurity practices. The internship strengthened my technical foundation, improved my problem-solving abilities, and increased my confidence in working with cybersecurity concepts. The knowledge and skills acquired during this internship will support my future academic studies and professional career in cybersecurity.

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Annexures

Internship Offer Letter

Bengaluru, Karnataka – 560095, India
internship@internspark.in
www.internspark.in
Date: 05/06/2026

Internship Offer Letter

Dear Prince kumar,

Candidate ID: IS-2026-9871

We are delighted to offer you the position of **Cybersecurity Intern** at **InternSpark**. Your academic background, skills, and overall enthusiasm clearly reflect your potential, and we believe that you will be an excellent addition to our internship program.

At InternSpark, we are committed to nurturing emerging talent by providing a structured, growth-oriented learning environment. Throughout your internship, you will engage in meaningful, hands-on assignments designed to develop your technical competence, problem-solving abilities, and professional confidence.

This internship follows a **task-based learning model**, enabling you to gain practical exposure through real-world projects relevant to your chosen domain. Each task has been designed to simulate actual industry work, helping you build a strong technical foundation and preparing you for future professional opportunities.

You will also receive access to curated learning resources, timely guidance from mentors, and support from our internship coordination team. We encourage you to maintain active communication, seek clarification whenever required, and approach each task with dedication and a willingness to learn.

Upon successful completion of all assigned tasks and adherence to internship guidelines, you will be awarded an official **Internship Completion Certificate**. Based on your performance and overall contribution, you may also become eligible for a **Letter of Recommendation (LOR)**, which can serve as a valuable credential in your future academic or professional pursuits.

During your internship journey, we expect you to demonstrate professionalism, discipline, and a positive learning attitude. We believe that your contributions will not only support your individual growth but will also align with InternSpark's vision of fostering innovation and excellence.

We are excited to welcome you to **InterSpark** and look forward to seeing you develop your skills, achieve your goals, and make the most of this opportunity.

We wish you great success ahead. Let's grow together!

Duration: 1 Month

Starting Date: 05/06/2026

Sincerely,

Founder & CEO

ISO 9001:2015

Internship Completion Certificate


internSpark

All India Council for Technical Education
AICTE

CERTIFICATE
OF COMPLETION

THIS CERTIFICATE IS PRESENTED TO

Prince Kumar

In recognition of his/her efforts and achievements in completing
1 Month internship in Cybersecurity.

Conducted From 5th June 2026 to 5th July 2026

ALFIDO TECH
RIDING THE WAVES OF TECHNOLOGICAL ADVANCEMENT

Candidate ID
IS-2026-9871

Scan to Verify Certificate Authenticity

ISO
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सत्यमेव जयते
MSME

Authorized Signatory
Issued by Alfido Tech – Registered Training Organization

Letter of Recommendation

internSpark

Bengaluru, Karnataka – 560095, India
internship@internspark.in
www.internspark.in
Date: 05/07/2026

Letter of Recommendation

To Whom It May Concern,

This is to formally recommend **Prince kumar** (Candidate ID: **IS-2026-9871**) for future academic or professional opportunities. He/She successfully completed a **1 Month** remote internship in the role of **Cybersecurity Intern** at **InternSpark** from **05/06/2026** to **05/07/2026**.

During this period, he/she worked on a series of **task-based assignments and real-world projects** relevant to the chosen domain. These tasks were designed to simulate practical industry scenarios, and the intern consistently demonstrated the ability to understand requirements, deliver quality work, and adhere to timelines in a remote working environment.

He/She exhibited an outstanding level of performance, consistently exceeding expectations throughout the internship. the intern showed a proactive approach to learning, regularly sought clarification when needed, and effectively incorporated feedback to improve the quality of deliverables.

In addition to technical growth, he/she displayed strong professional values such as discipline, responsibility, and clear communication. He/She coordinated well with mentors and peers, maintained a positive attitude, and contributed meaningfully to the ongoing activities of the internship program.

Based on his/her overall performance, we can confidently state that the candidate has developed a solid foundation in **Cybersecurity** and is well-prepared to take up future roles in the technology and professional ecosystem. We believe that He/She will be an asset to any organization or institution that may offer he/she an opportunity.

This Letter of Recommendation is being issued upon the request of the candidate to support his/her future academic, internship, or job applications.

If you require any further information or wish to verify the details of this internship, please feel free to contact us at **internship@internspark.in**.

We extend our best wishes to Prince kumar for a successful career ahead.

Sincerely,

Founder & CEO

ISO 9001:2015

MSME

Internship Report Card / Performance Evaluation

internSpark

Prince kumar
Cybersecurity
Candidate ID: **IS-2026-9871**
Period: **05/06/2026 - 05/07/2026**
Cert/Link: View

Report Card

Overall Percentage: 91.65%

Grade: A+

Outstanding Performance

Maintor/Supervisor - Akash Dubey (akash.dubey@internspark.in)

Week 1 — 05/06/2026 to 11/06/2026

Date	Topic / Activity	Marks	Remarks
05/06/2026	Introduction to Cybersecurity	89	Excellent
06/06/2026	Networking Basics	89	Excellent
07/06/2026	Types of Threats	96	Excellent
08/06/2026	Encryption Techniques	93	Excellent
09/06/2026	Vulnerabilities	95	Excellent
10/06/2026	Penetration Testing	93	Excellent
11/06/2026	Firewalls	90	Excellent

Week 2 — 12/06/2026 to 18/06/2026

Date	Topic / Activity	Marks	Remarks
12/06/2026	Ethical Hacking	86	Excellent
13/06/2026	Incident Response	91	Excellent
14/06/2026	SOC Operations	87	Excellent
15/06/2026	Cyber Laws & Compliance	91	Excellent
16/06/2026	Introduction to Cybersecurity	88	Excellent
17/06/2026	Networking Basics	85	Excellent
18/06/2026	Types of Threats	92	Excellent

Week 3 — 19/06/2026 to 25/06/2026

Date	Topic / Activity	Marks	Remarks
19/06/2026	Encryption Techniques	97	Excellent
20/06/2026	Vulnerabilities	92	Excellent
21/06/2026	Penetration Testing	98	Excellent
22/06/2026	Firewalls	97	Excellent
23/06/2026	Ethical Hacking	89	Excellent
24/06/2026	Incident Response	96	Excellent
25/06/2026	SOC Operations	89	Excellent

Week 4 — 26/06/2026 to 02/07/2026

Date	Topic / Activity	Marks	Remarks
26/06/2026	Cyber Laws & Compliance	85	Excellent
27/06/2026	Introduction to Cybersecurity	98	Excellent
28/06/2026	Networking Basics	91	Excellent
29/06/2026	Types of Threats	85	Excellent
30/06/2026	Encryption Techniques	94	Excellent
01/07/2026	Vulnerabilities	86	Excellent
02/07/2026	Penetration Testing	91	Excellent

Week 5 — 03/07/2026 to 05/07/2026

Date	Topic / Activity	Marks	Remarks
03/07/2026	Firewalls	98	Excellent
04/07/2026	Ethical Hacking	91	Excellent
05/07/2026	Incident Response	99	Excellent

Akash Dubey
Maintor/Supervisor

Scan to Verify

Generated on: 08/07/2026